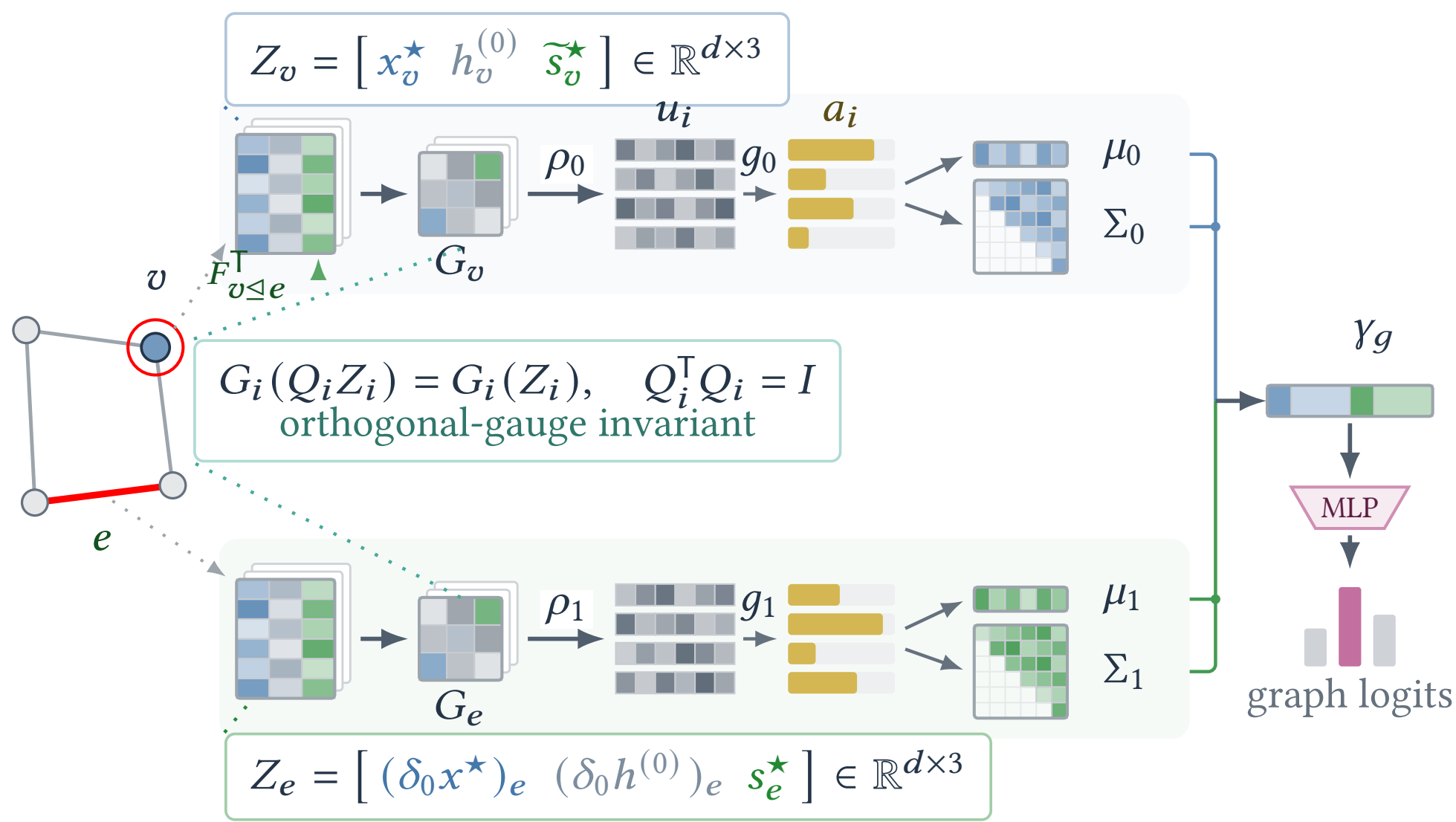

$Z_v = \left[x_v^\star \;\; h_v^{(0)} \;\; \widetilde{s_v^\star} \right] \in \mathbb{R}^{d\times 3}$
u_i
a_i
ρ_0
g_0
μ_0
Σ_0
G_v
v
$F_{v\trianglelefteq e}^\top$
$G_i(Q_i Z_i) = G_i(Z_i), \quad Q_i^\top Q_i = I$
orthogonal-gauge invariant
e
ρ_1
g_1
μ_1
Σ_1
G_e
γ_g
MLP
graph logits
$Z_e = \left[(\delta_0 x^\star)_e \;\; (\delta_0 h^{(0)})_e \;\; s_e^\star \right] \in \mathbb{R}^{d\times 3}$